OS

Home assignment - 4

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1) Device management Techniques for I/O optimization.

a) Buffering :

-> Temporary storage area in main memory.

b) Caching :

-> stores frequently accessed data in faster memory (cache).

c) Spooling :

-> Jobs are placed in a queue and executed sequentially.

d) I/O scheduling

-> OS decides the order of servicing I/O requests.

algorithms

-> FCFS (first come first serve)

-> SSTF (Shortest seek Time first).

-> SCAN (Elevator algorithm)

-> C- SCAN.

2) RAID levels

* RAID 0 - striping

Disk : A_1 A_3 A_5

Disk : A_2 A_4 A_6

-) Data split across disks

* RAID 1 - mirroring

Disk 1 : A_1 A_2 A_3

Disk 2 : A_1 A_2 A_3

-) Exact copy on another Disk.

* RAID 2 - Bit level striping + Hamming code

-) Rarely used.

-) uses error correction codes

* RAID 3 - Byte - level striping + parity Disk.

Disk 1 : A_1 A_4

Disk 2 : A_2 A_5

Disk 3 : A_3 A_6

Disk 4 : parity.

3) Internal structure of Hard Disk Drive

components

-> platters : magenetic disks storing data.

-> spindle : Rotates.

-> Read/write Head - Reads/writes data.

-> Actuator Arm -> moves head.

-> Tracks -> circular paths.

-> sectors -> smallest unit of storage.

-> cylinder -> Tracks across platters.

4) Distributed systems

-> A system where multiple computers work together as a single system.

Architecture Types

1) client server

2) peer to peer

3) Distributed database systems

4) microservices Architecture.

5) Data integrity and protection mechanisms

a) access control

b) encryption

c) Backup and Recovery

d) file and permissions.

e) Journaling

f) checksums.

6) file system implementation.

a) file allocation methods

b) contiguous allocation.

2) linked allocation.

3) Indexed allocation.

4) free space management

5) Directory structure.